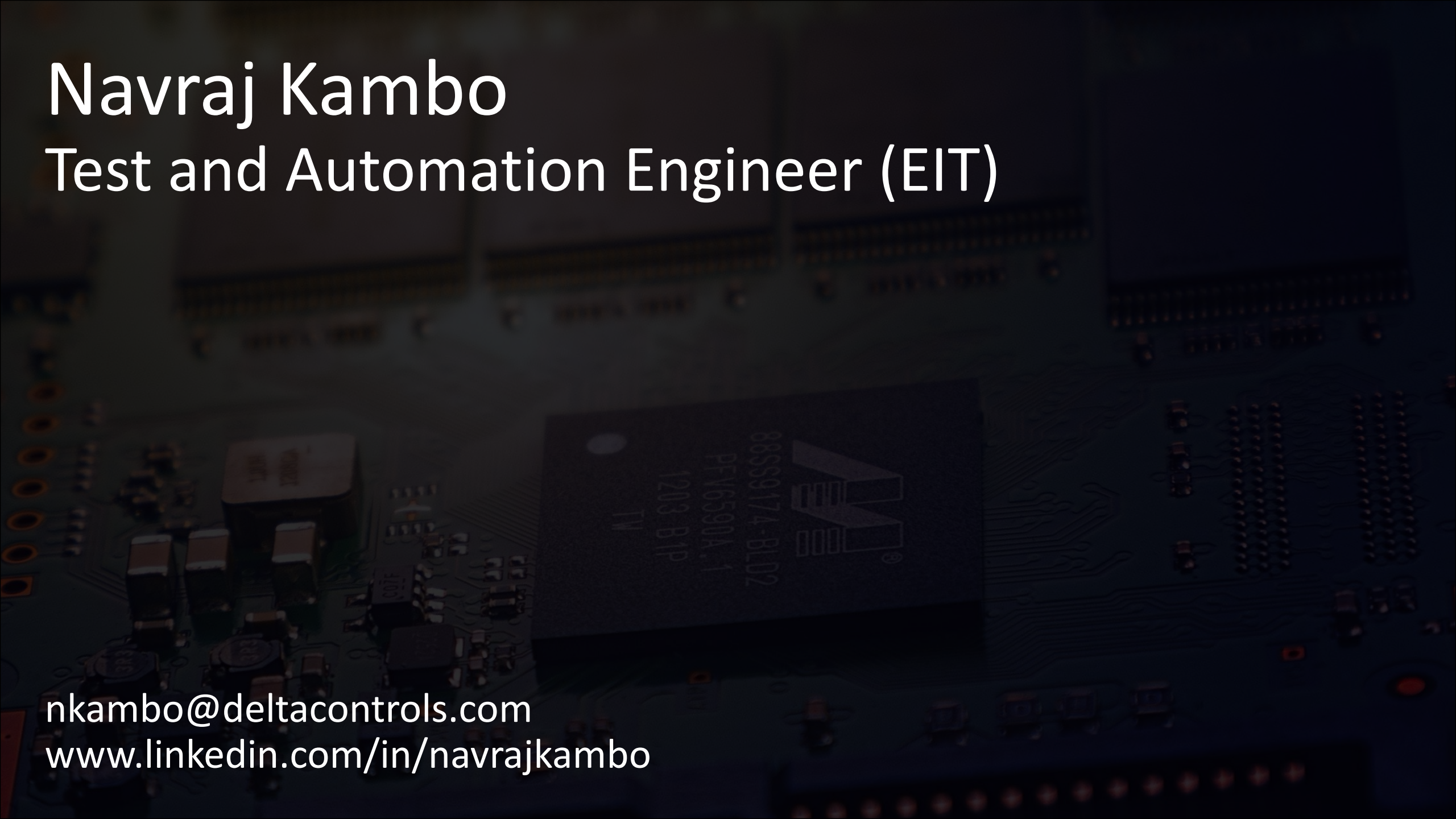




Navraj Kambo

Test and Automation Engineer (EIT)

nkambo@deltacontrols.com

www.linkedin.com/in/navrajkambo

Navraj Kambo | Overview

- 3 About Me
- 4 Digital Speech Decoding (WTS Wireless Technical Services Inc. / FortisBC)
- 5 NDART (British Columbia Institute of Technology)
- 6 Autonomous Vineyard Harvesting (British Columbia Institute of Technology SoE)
- 7 Apex Motion Control
- 8 Packaging Line(Apex Motion Control)
- 9 DecoBot (Apex Motion Control)
- 10 Wellons Canada
- 11 Continuous Drying System (Wellons Canada)
- 12 Delta Intelligent Building Technologies (Canada) Inc.
- 13 RED 5 Edge (China) (DIBT)
- 14 Test & Automation (DIBT)



Navraj Kambo | About Me

I'm an electrical engineer who graduated in 2019. I have experience in robotics, automation, and electronics. I enjoy collaborating with multi-disciplinary teams. Outside of work I enjoy playing sports.

Education



**BEng Electrical Engineering
Research Intern (4 months)**



MEng ECE

Experience



4-month co-op



Design Engineer



Electrical Engineer EIT



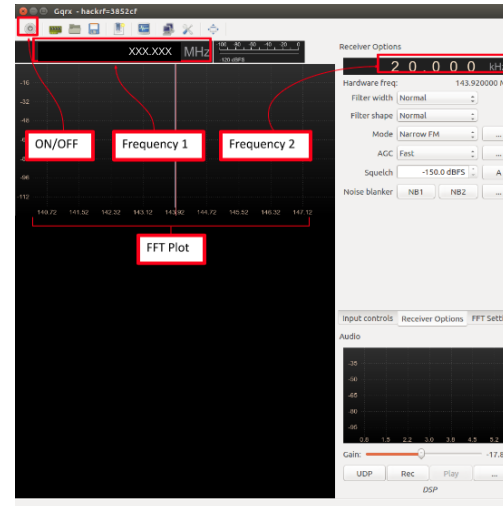
Delta Intelligent Building Technologies

Test & Automation Engineer

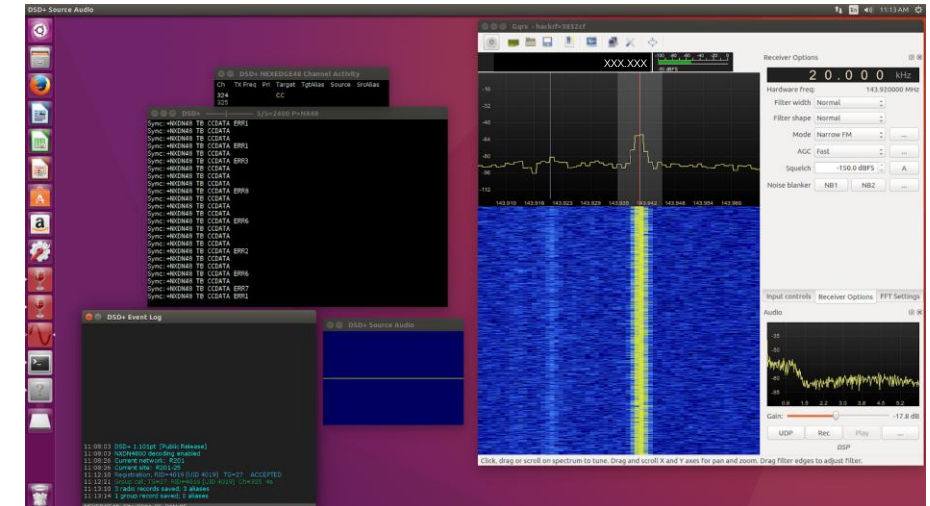
Navraj Kambo | Digital Speech Decoding

```
wts-nav@KRYPTONITE: ~  
wts-nav@KRYPTONITE:~$ hackrf_info  
Found HackRF board 0:  
USB descriptor string: 000000000000000909864c8383852cf  
Board ID Number: 2 (HackRF One)  
Firmware Version: 2015.07.2  
Part ID Number: 0xa000cb3c 0x00594f4f  
Serial Number: 0x00000000 0x00000000 0x909864c8 0x383852cf  
wts-nav@KRYPTONITE:~$
```

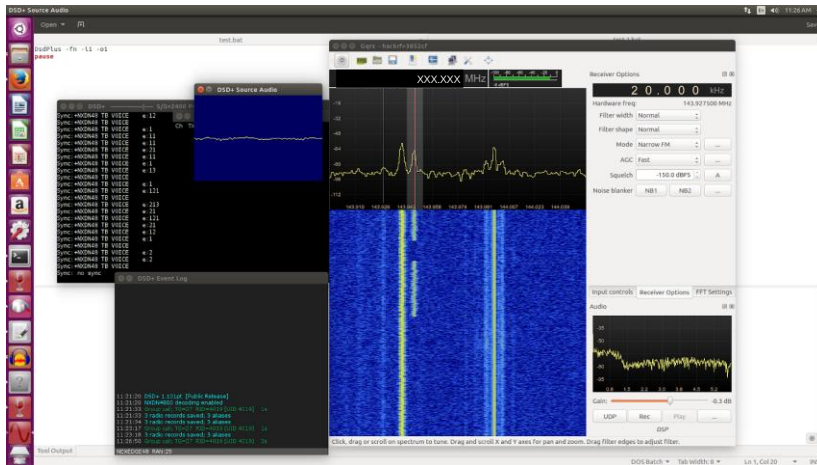
HackRF in Ubuntu



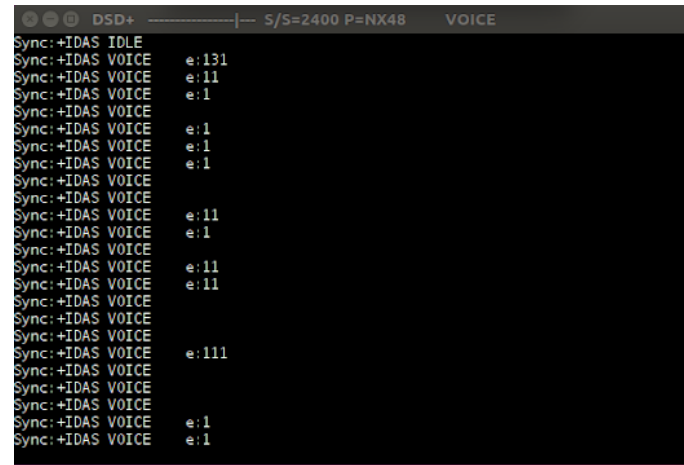
GQRX SDR



Decoding Control Channel (Carrier)



Decoding Traffic Channel (Side-band)

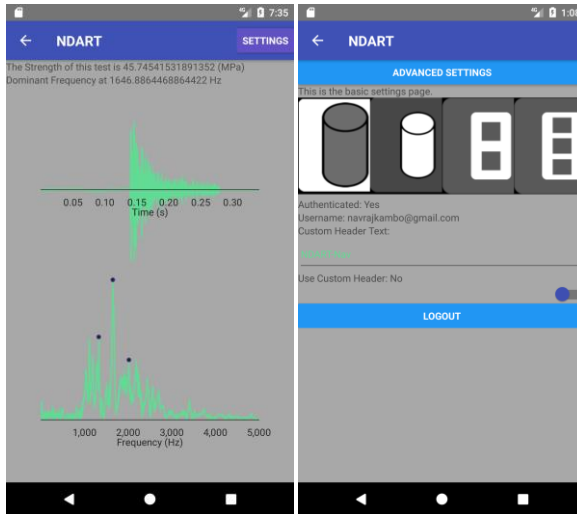


Sync Window on Traffic Channel

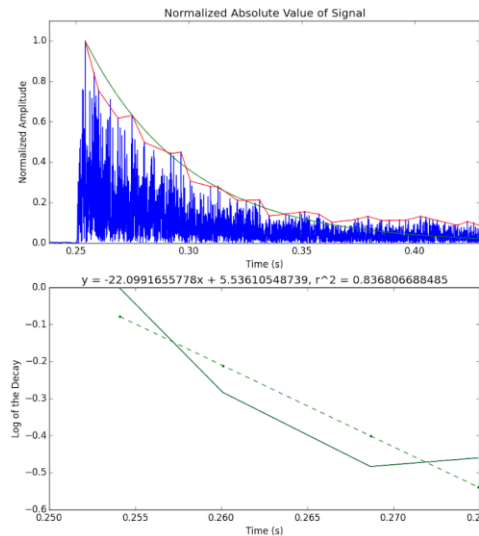


HackRF One

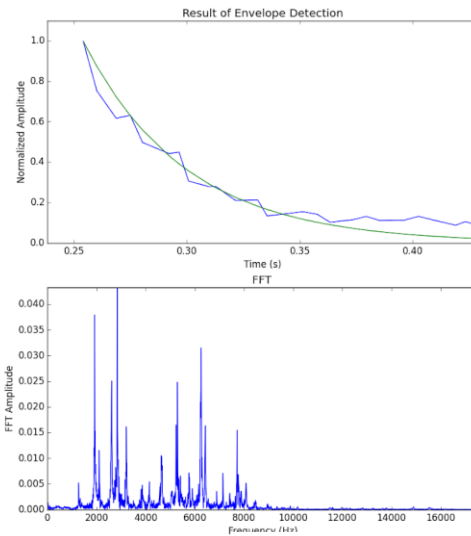
Navraj Kambo | NDART



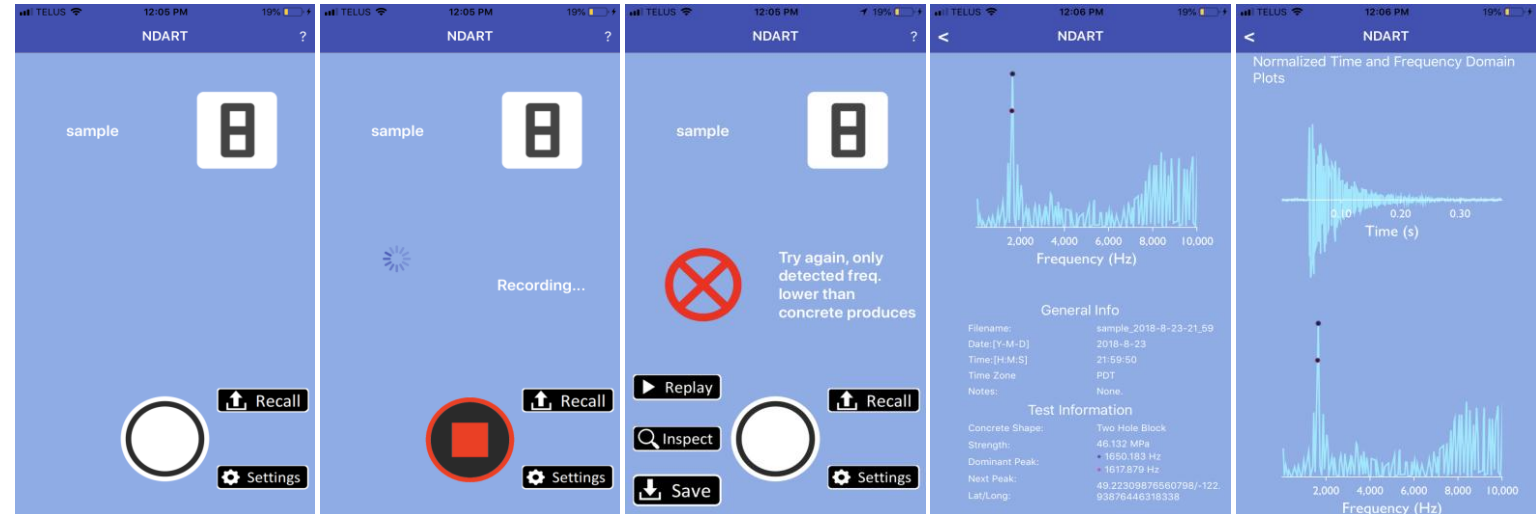
NDART app on Android



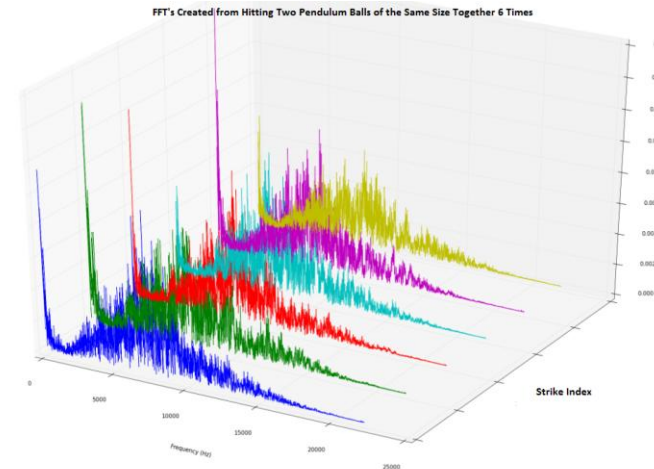
Strike Decay Plot



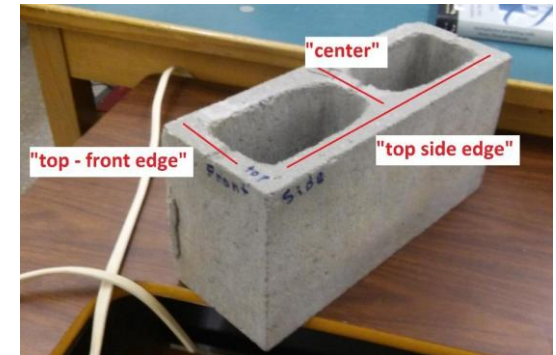
$$N(t) = N_0 e^{-\lambda t}$$



NDART app on iOS



FFTs at Different Locations

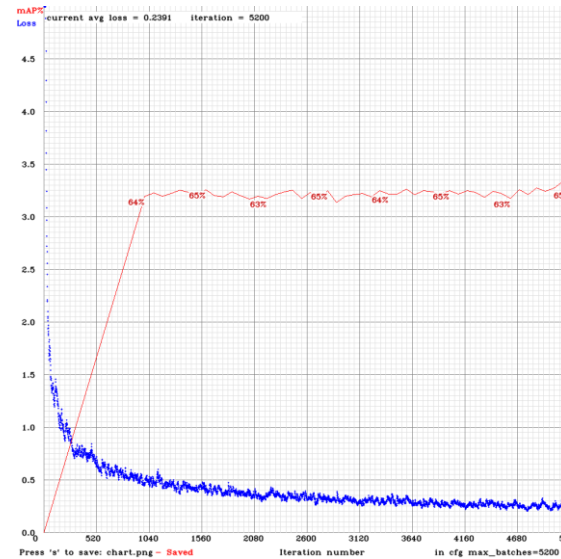


Concrete Block

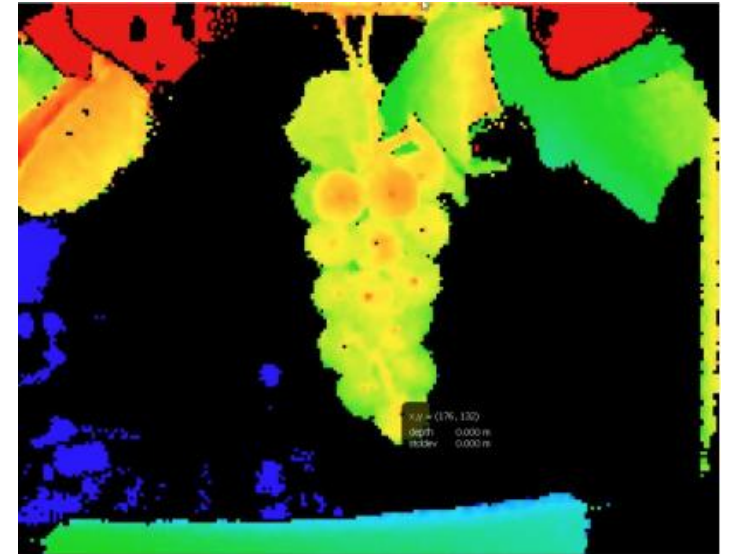
Navraj Kambo | Autonomous Vineyard Harvesting



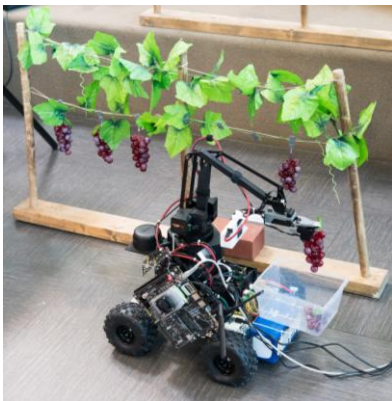
Project Expo



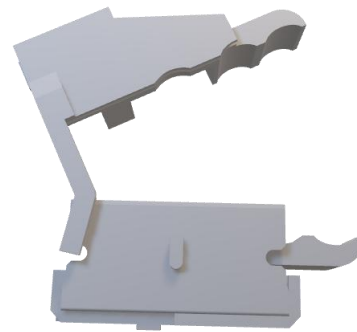
ML Dataset Training



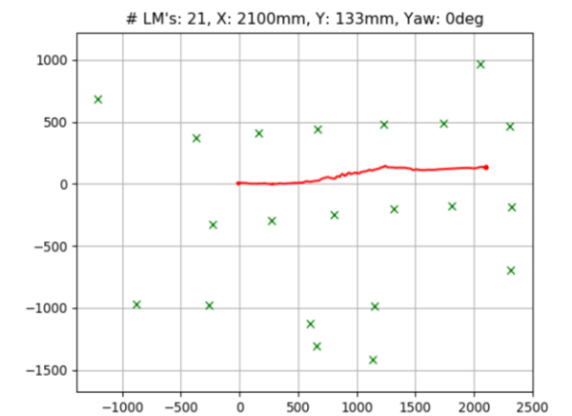
Grape Cluster TOF Image



Vinebot

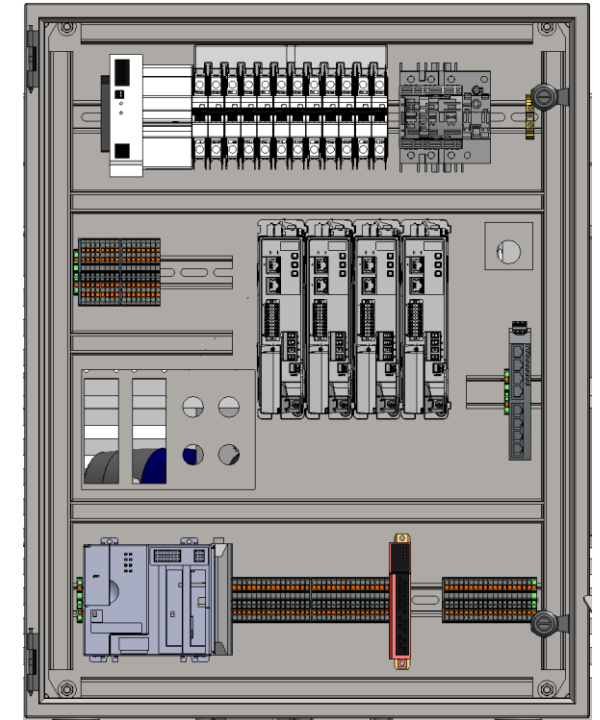
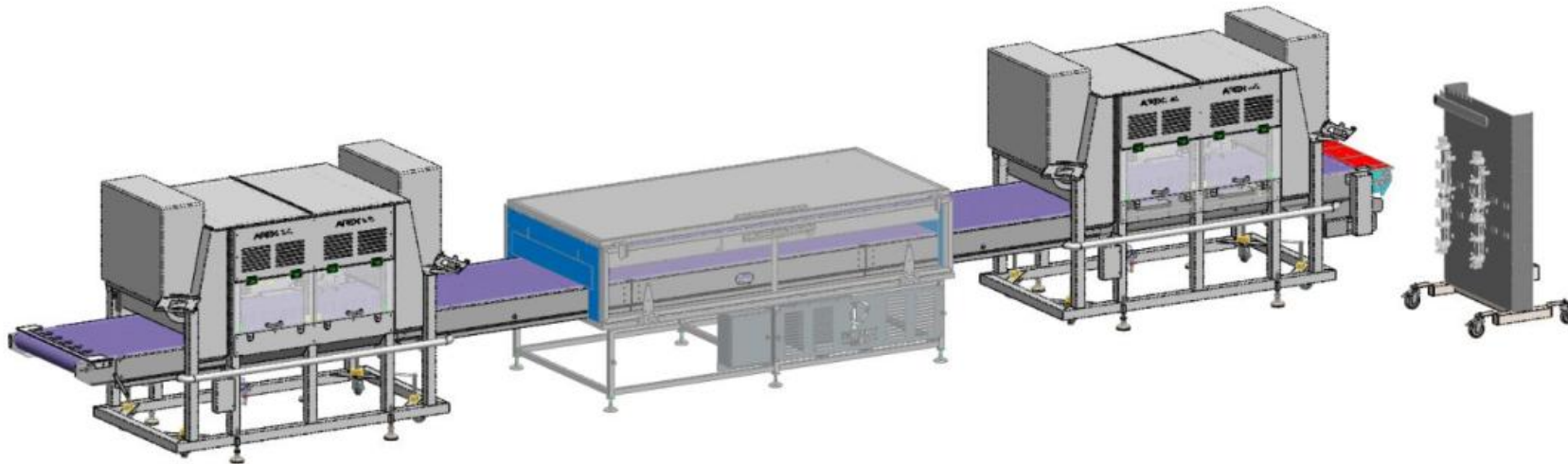


Vinebot Stem Cutter

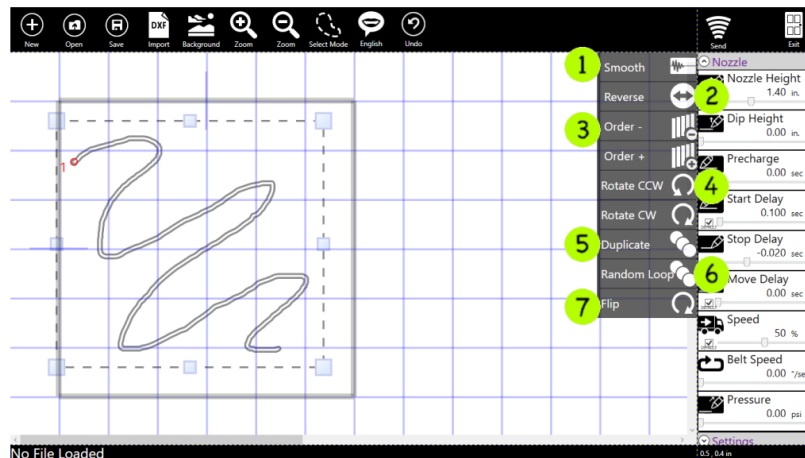


SLAM Path Planning

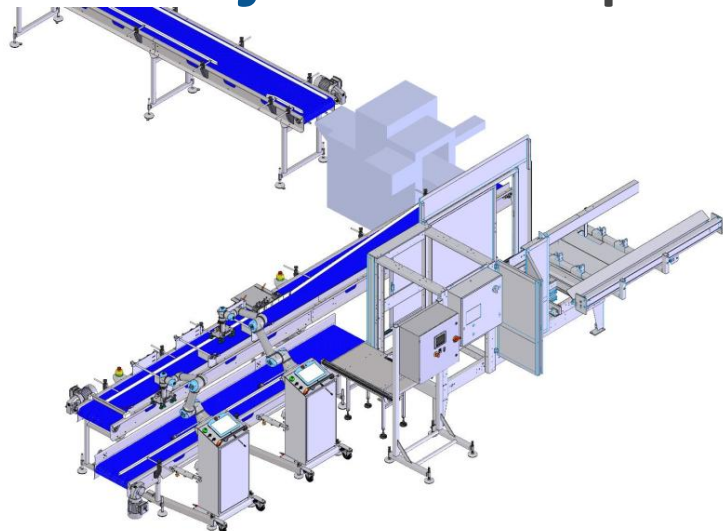
Navraj Kambo | Apex Motion Control - Design Engineer



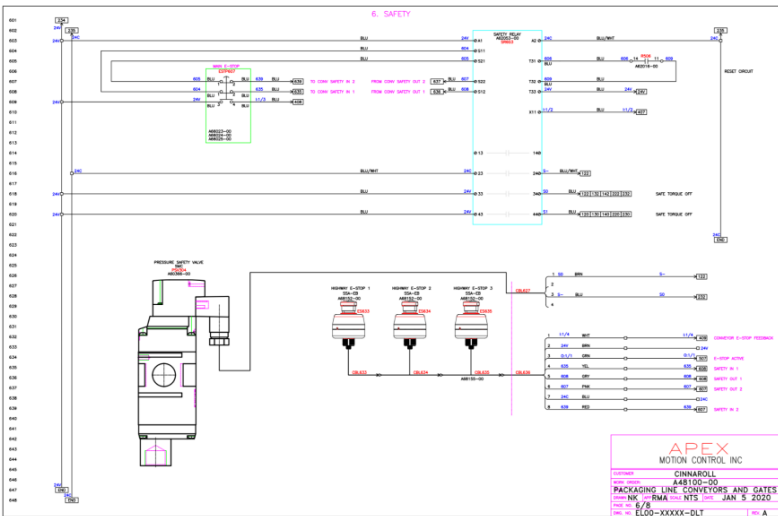
Robotics and Automation (Food and Beverage Industry)



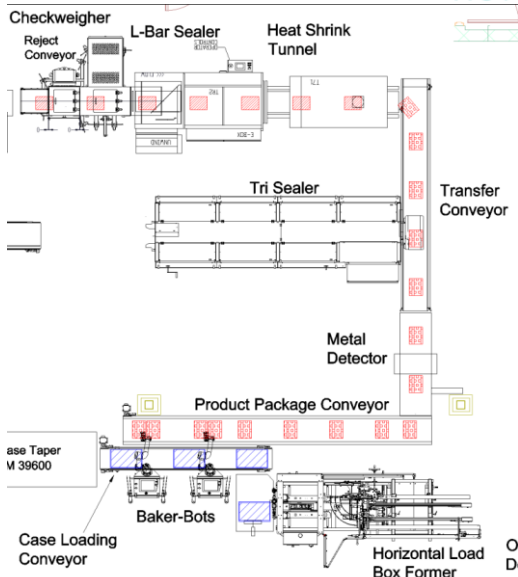
Navraj Kambo | Packaging Line



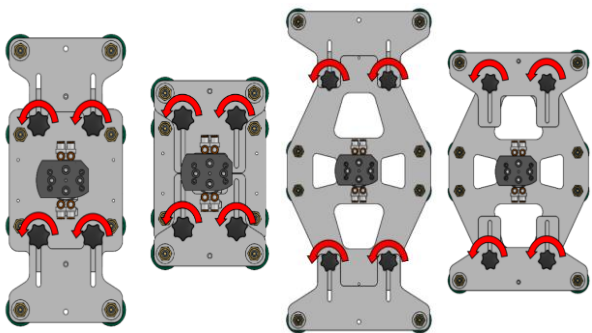
SolidWorks Assembly



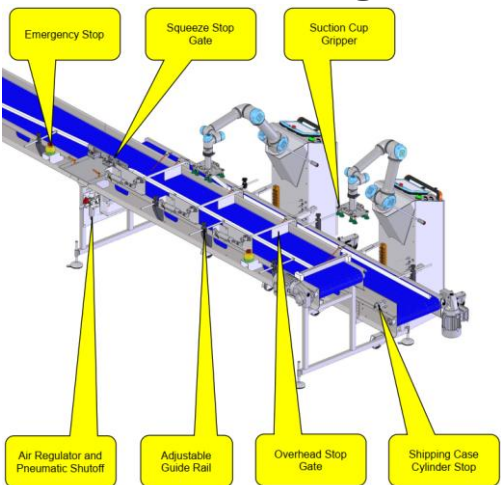
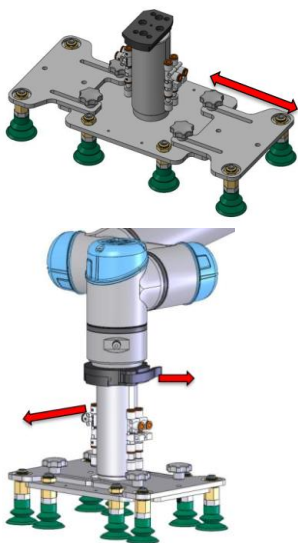
Electrical Design



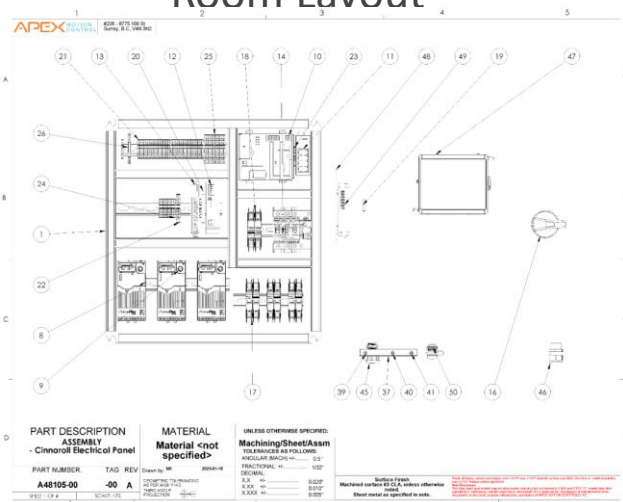
Room Layout



Adjustable Tooling

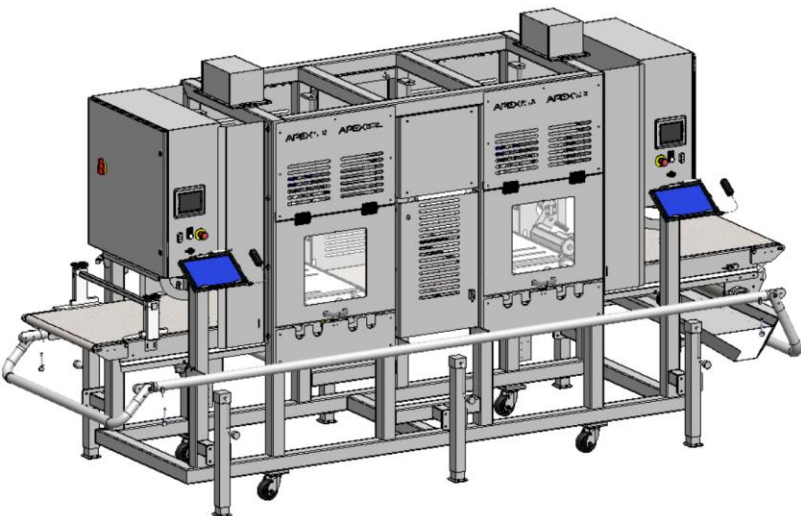


Pick & Place Cell

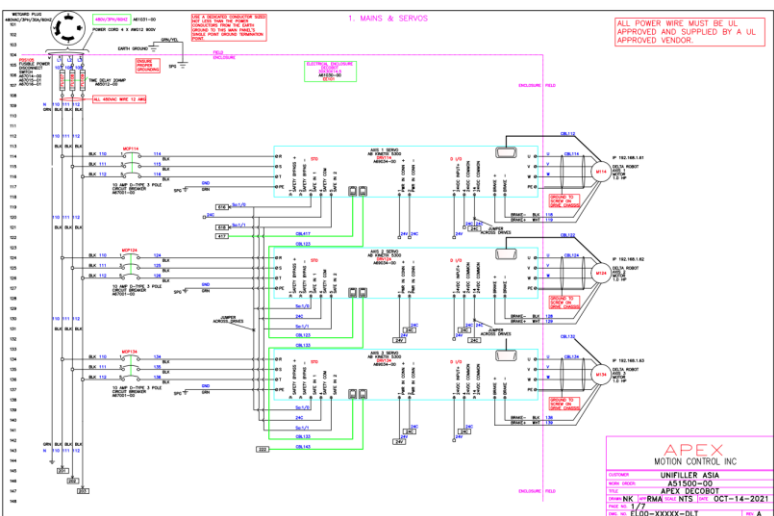


Electrical Panel Layout

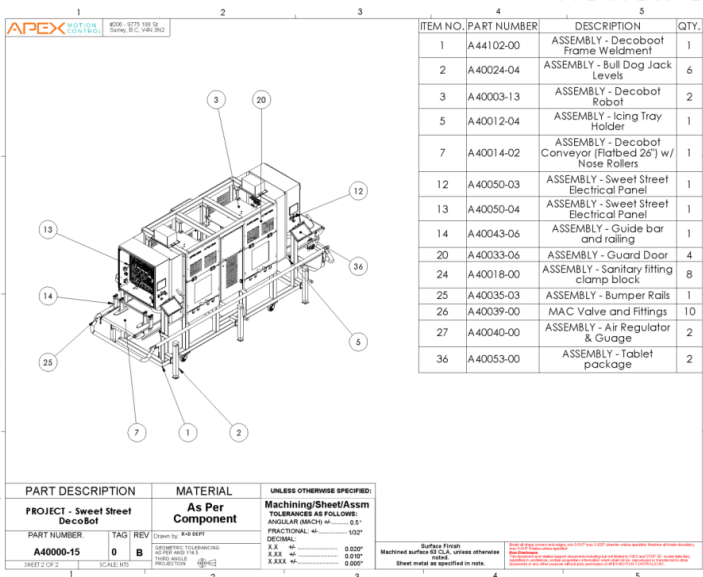
Navraj Kambo | DecoBot



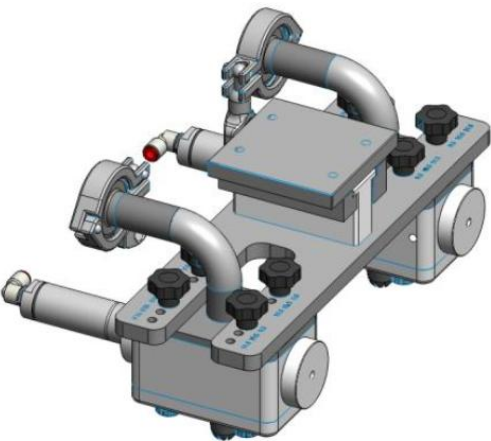
SolidWorks Assembly (Dual DecoBot)



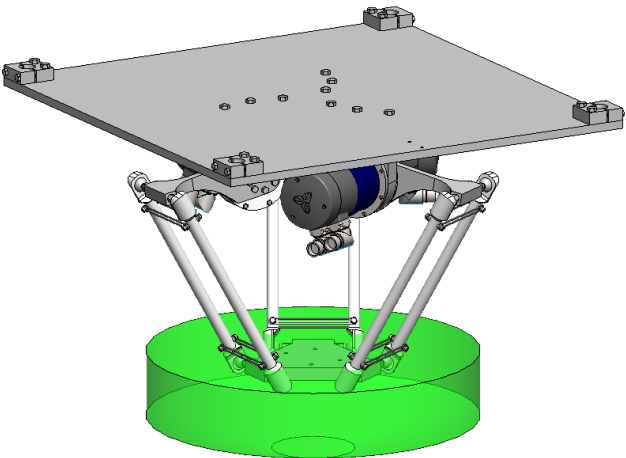
Electrical Design



Machine Drawing



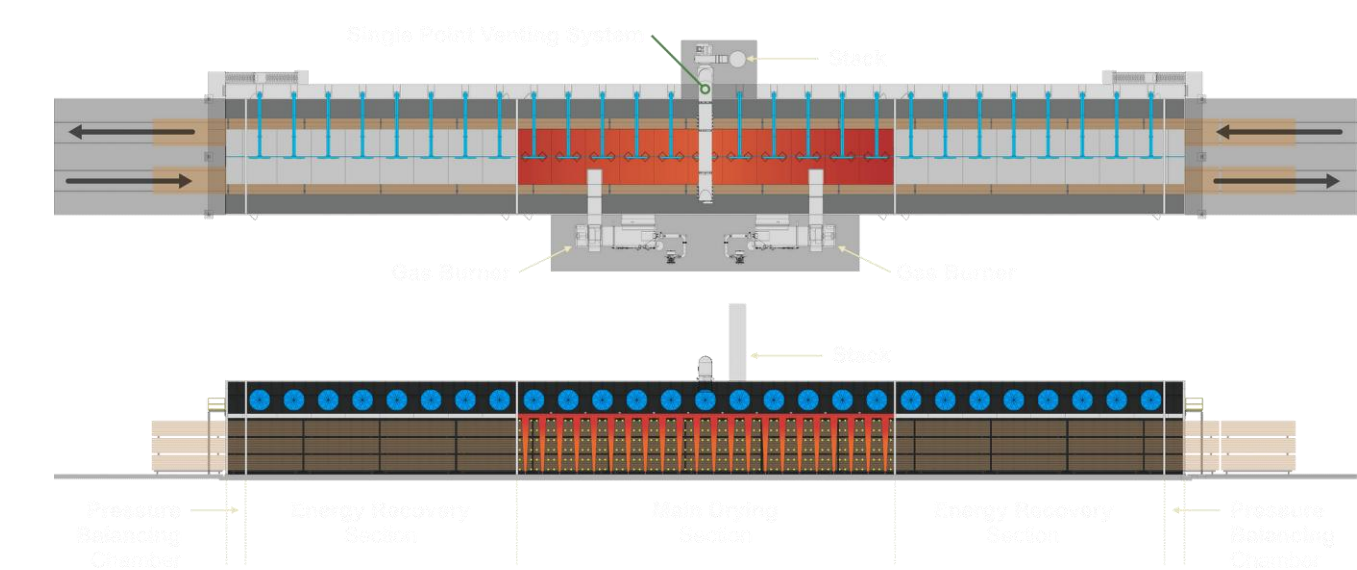
Robot Decorating Nozzle



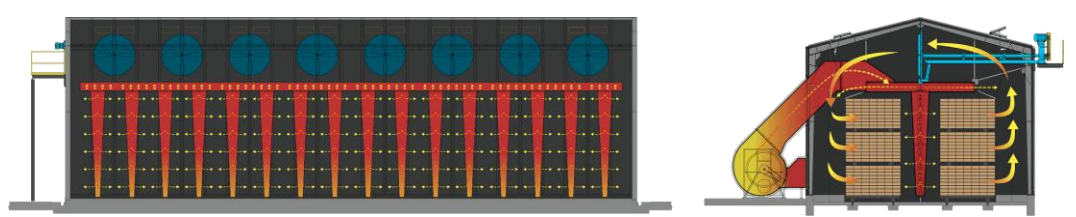
Delta Robot



Actual Robot

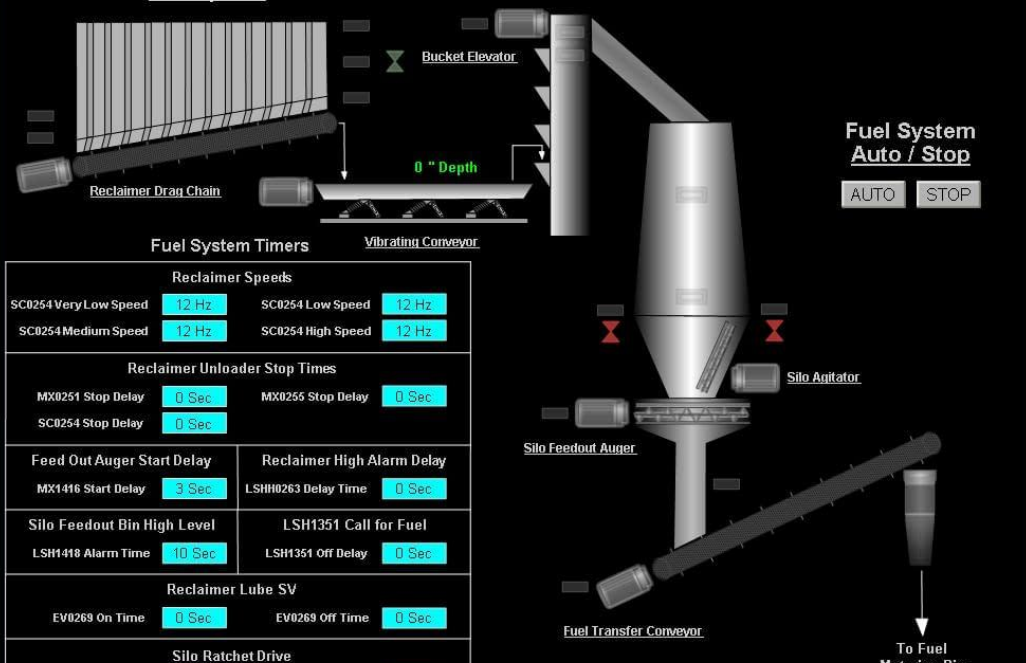


Wood Drying Kilns and Energy Systems



FuelSystem

Fuel System



Fuel System Auto / Stop
[AUTO] [STOP]

Fuel System Timers

Reclaimer Speeds	
SC0254 Very Low Speed	12 Hz
SC0254 Medium Speed	12 Hz
SC0254 Low Speed	12 Hz
SC0254 High Speed	12 Hz

Reclaimer Unloader Stop Times	
MX0251 Stop Delay	0 Sec
MX0255 Stop Delay	0 Sec
SC0254 Stop Delay	0 Sec

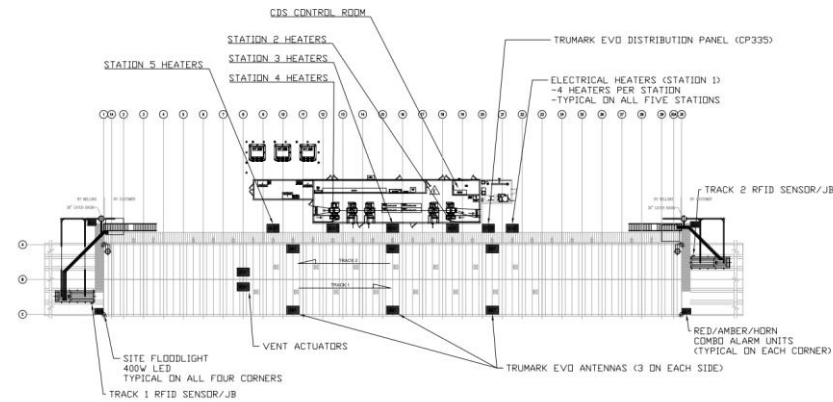
Feed Out Auger Start Delay	MX1416 Start Delay	3 Sec
Reclaimer High Alarm Delay	LSH0263 Delay Time	0 Sec
Silo Feedout Bin High Level	LSH1418 Alarm Time	10 Sec
LSH1351 Call for Fuel	LSH1351 Off Delay	0 Sec

Reclaimer Lube SV	
EV0269 On Time	0 Sec
EV0269 Off Time	0 Sec

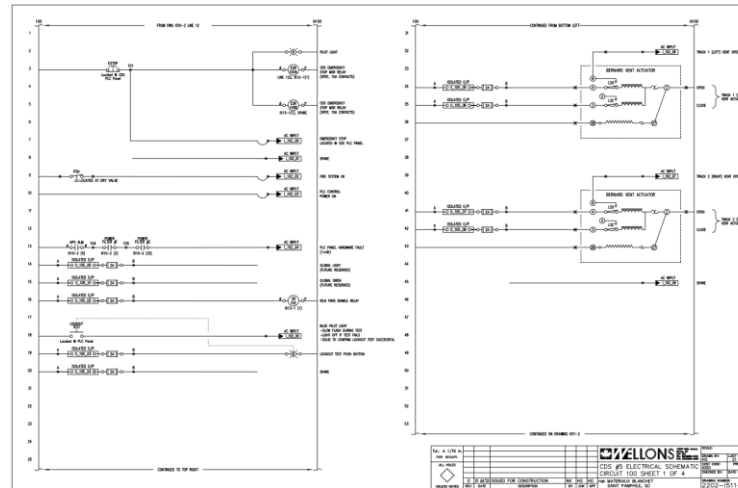
Silo Ratchet Drive	
EV1415A Ext Time	10 Sec
EV1415A Ret Time	10 Sec
EV1415B Ext Time	10 Sec
EV1415B Ret Time	10 Sec



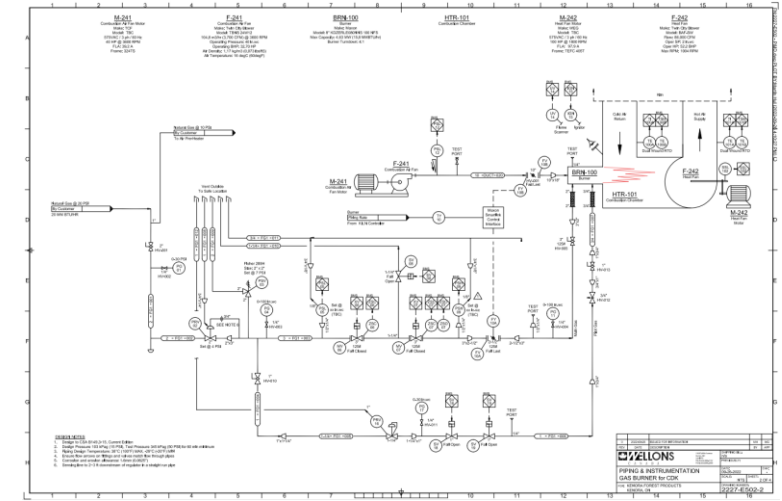
Navraj Kambo | Continuous Drying System



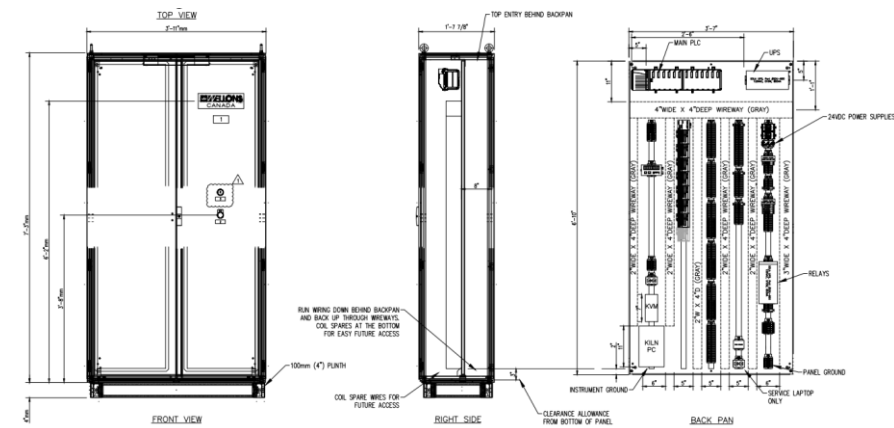
Kiln Layout Drawings



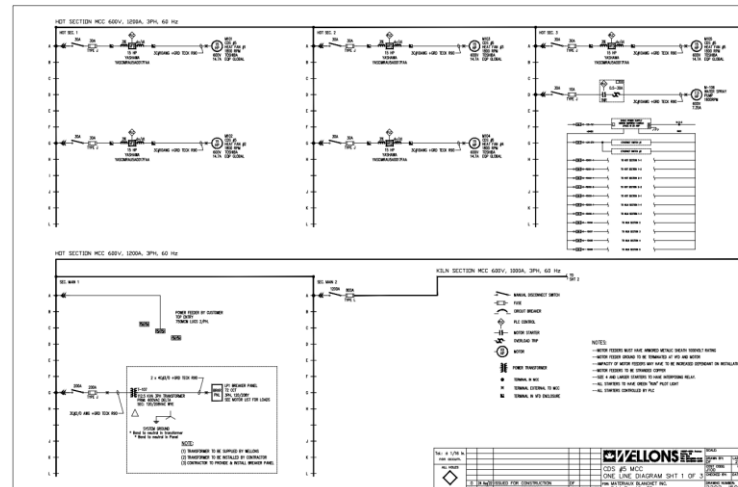
Control Schematics



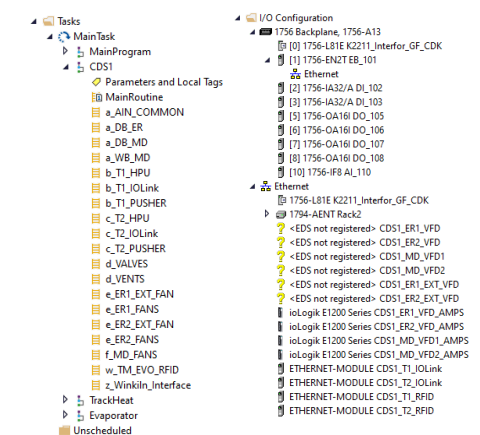
Gas Train P&IDs



Control Panel Design

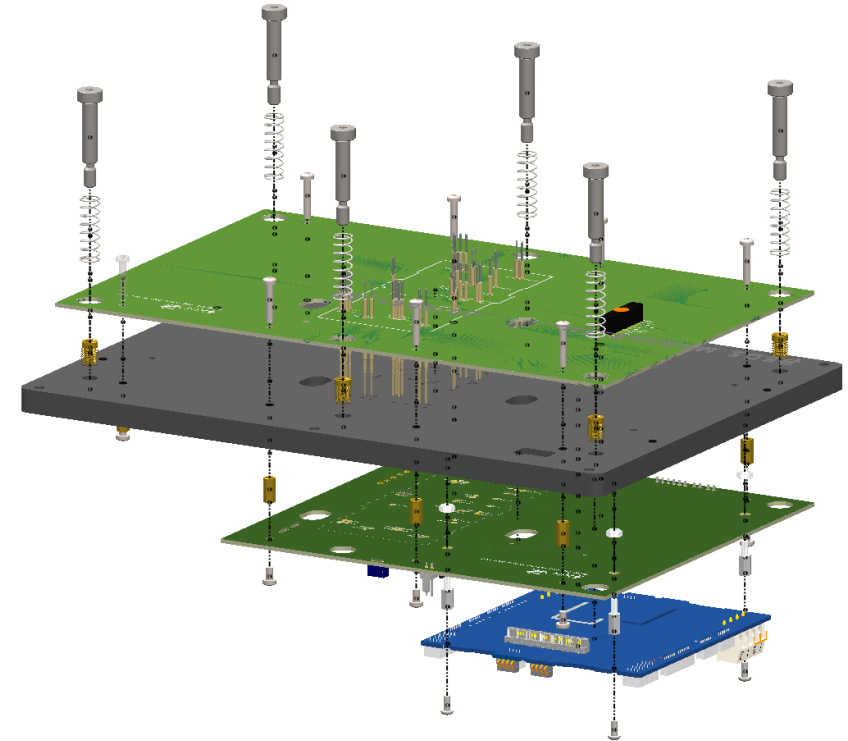
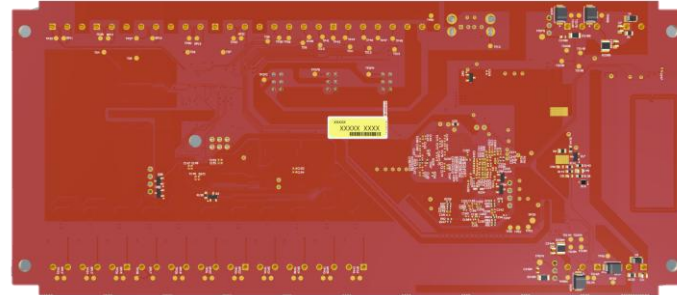
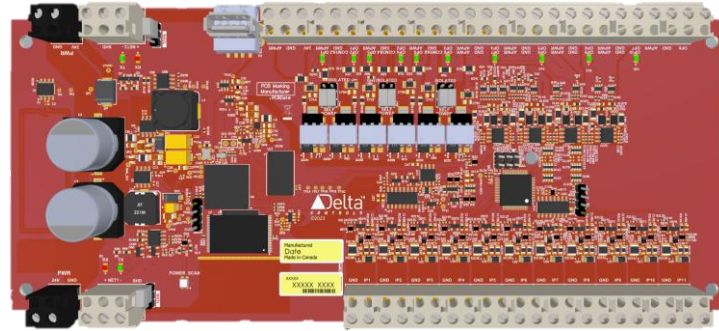
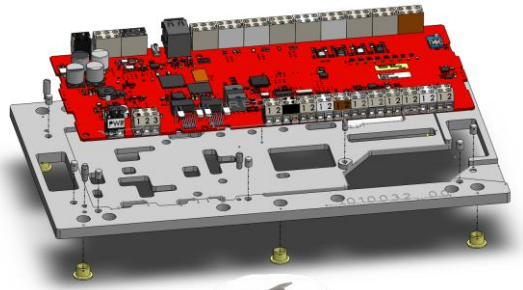


MCC Layout

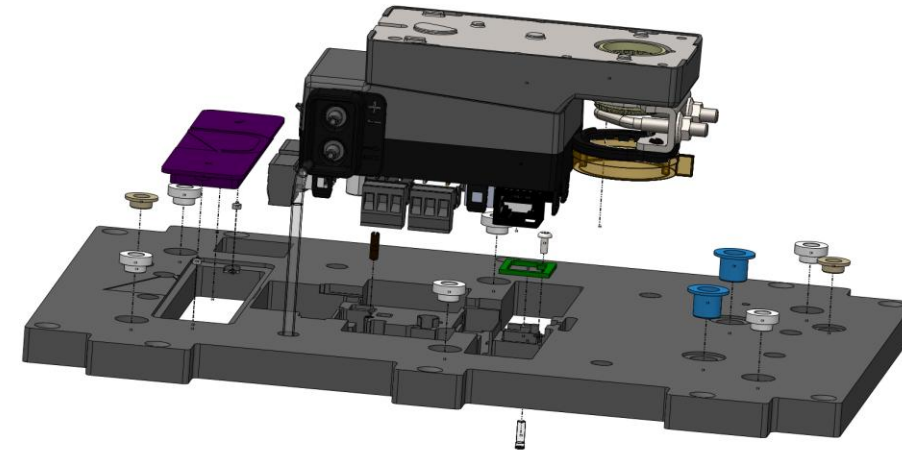
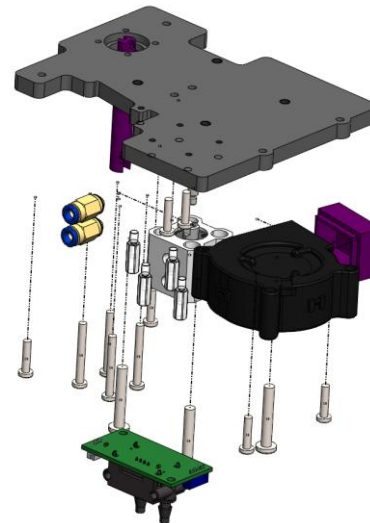
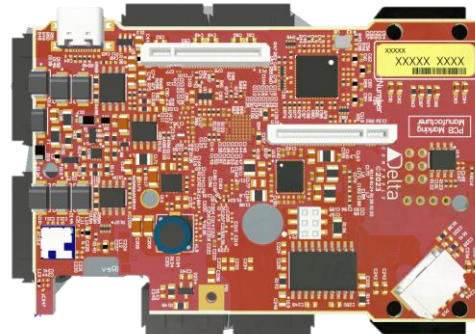
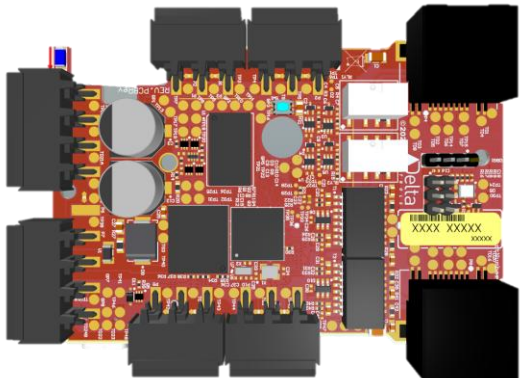


PLC Programming

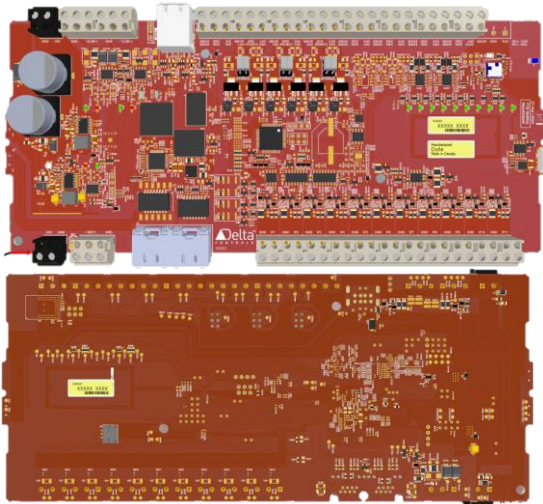
Navraj Kambo | DIBT – TAG Engineer



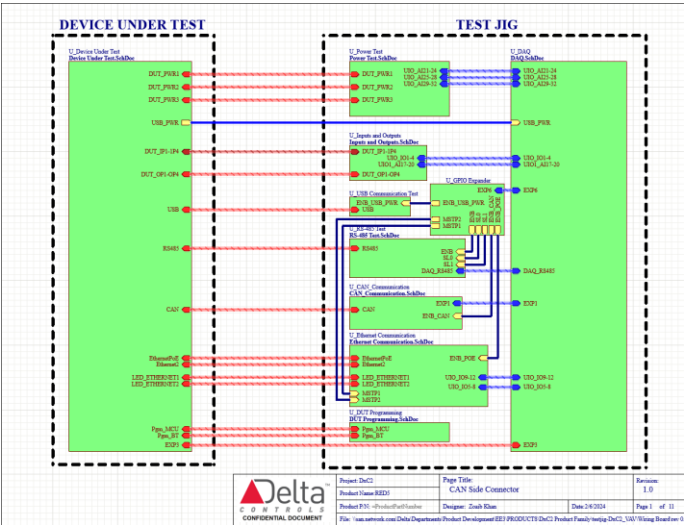
HVAC Controllers and Smart Thermostats (Testing Group)



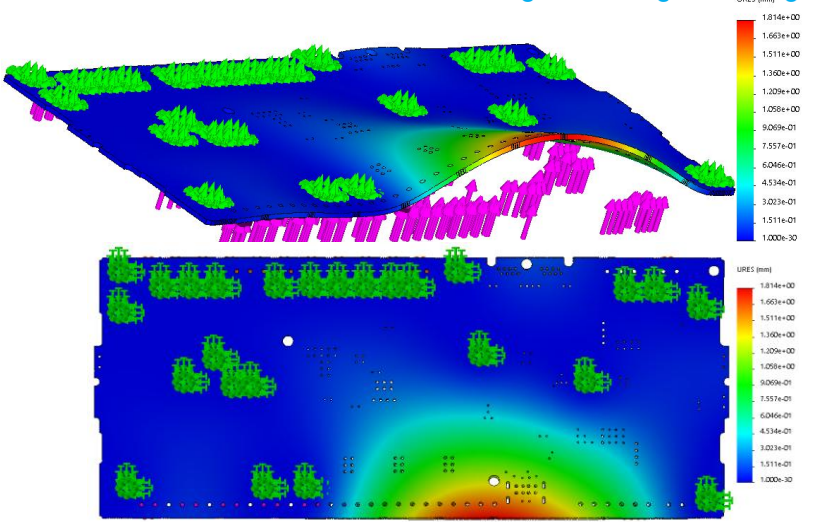
Navraj Kambo | RED 5 Edge (China)



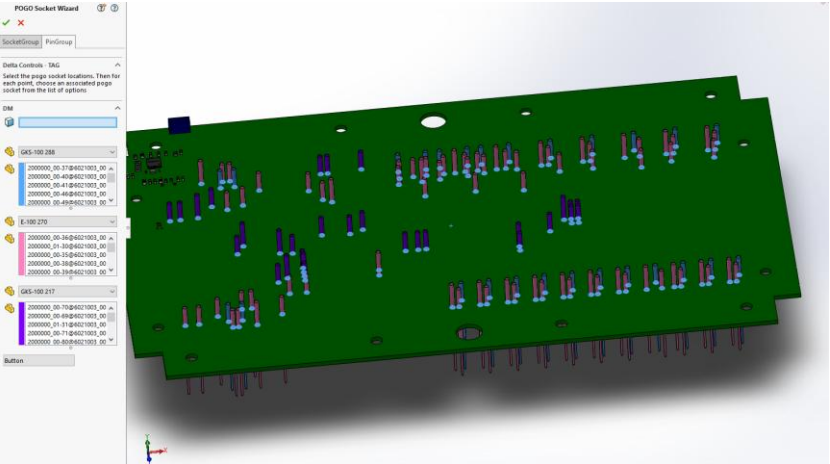
Hardware Review



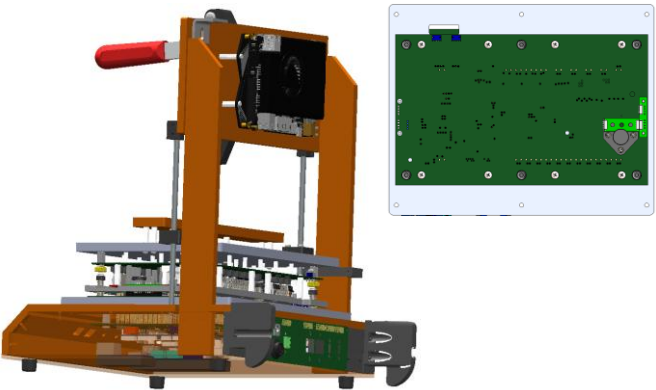
Test Jig Schematic Review



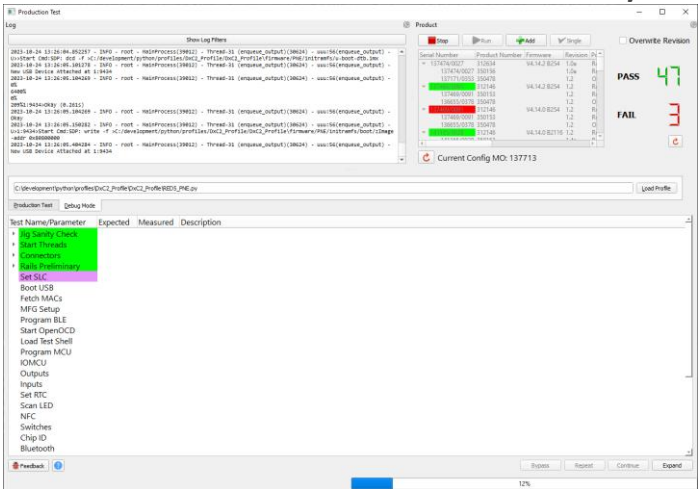
POGO Pin Static Analysis



POGO Pin/Socket Selection

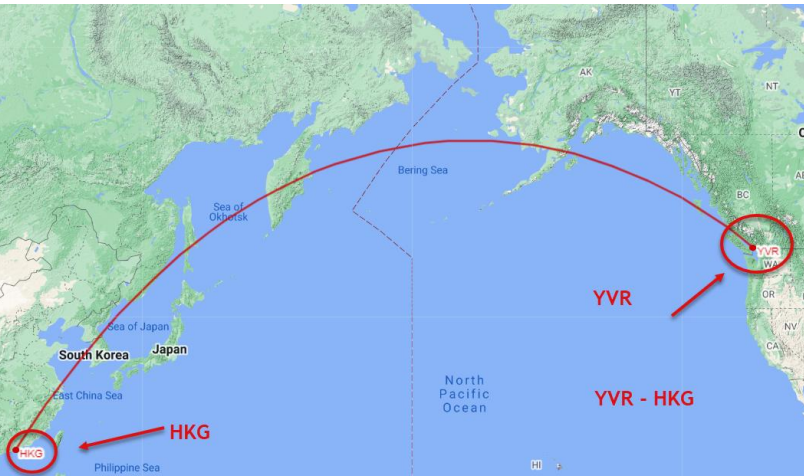


Mechanical Design

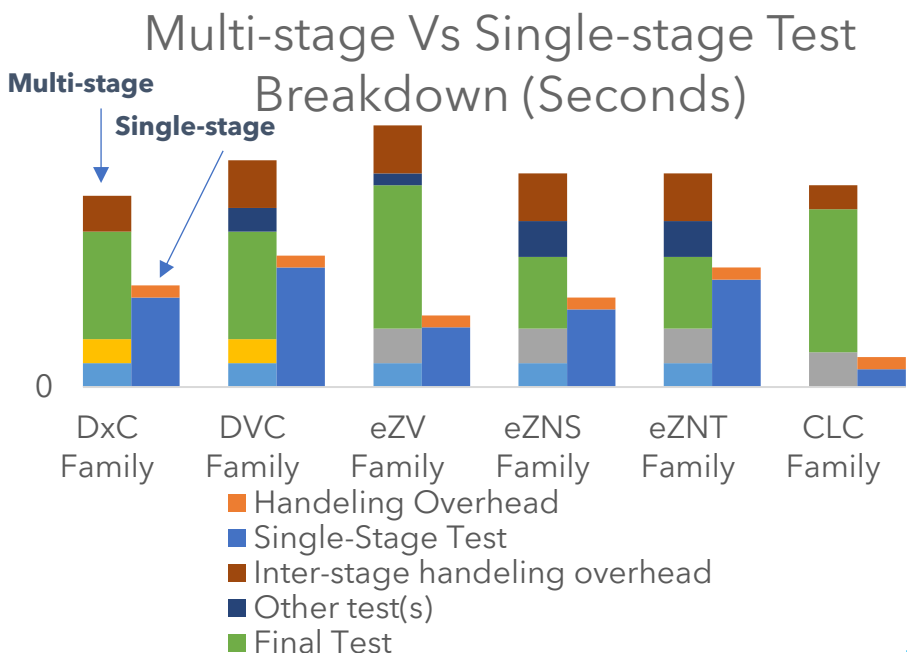


Test Firmware and Framework(UI)

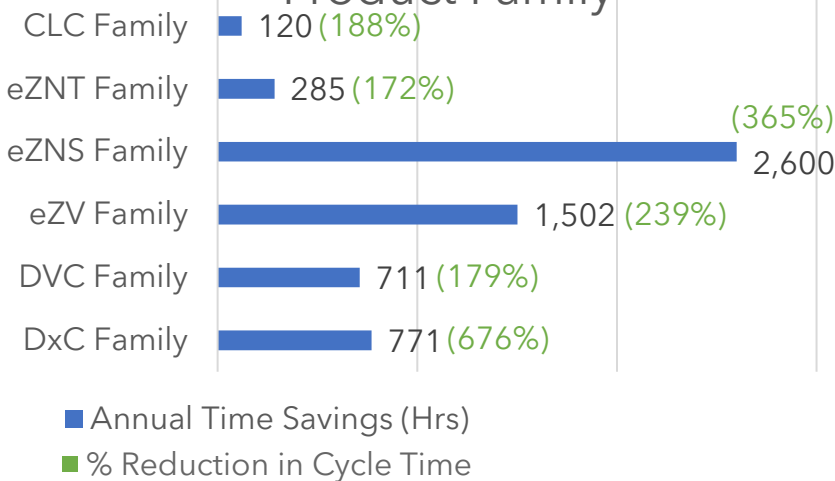
Navraj Kambo | RED 5 Edge (China)



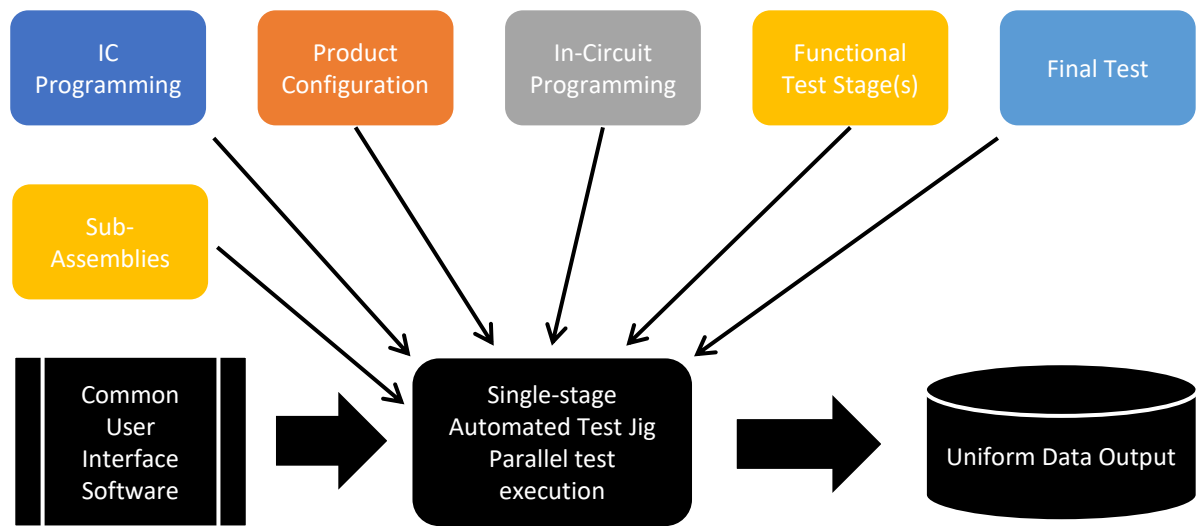
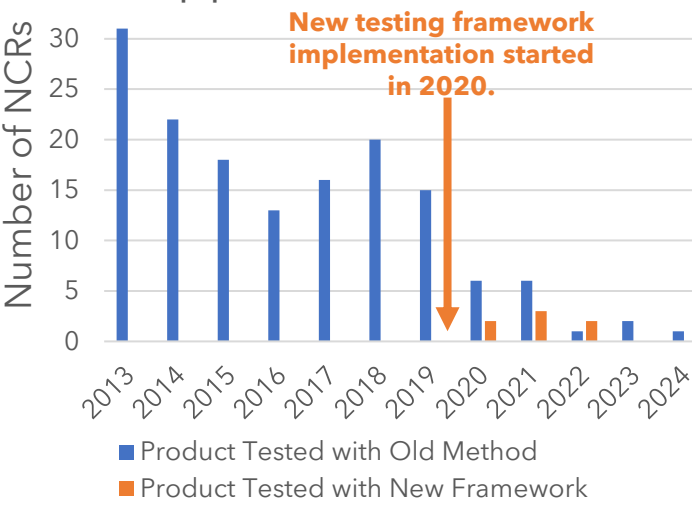
Navraj Kambo | Test and Automation Group



Annual Time Savings and Cycle Time Reduction by Product Family

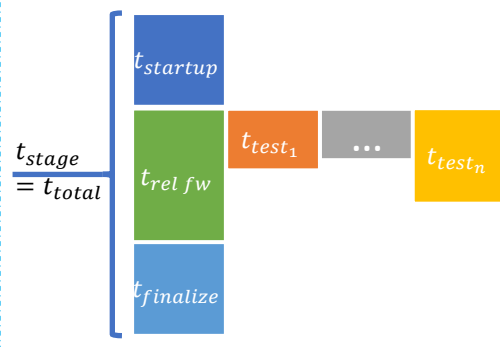


DIBT (Canada) Inc. Non-Conformance Reports applicable to Test



The DIBT (Canada) Inc. Test and Automation team developed a flexible and efficient framework (software, firmware, hardware, and data analysis) that integrates multiple production stages into a single streamlined process.

- Key benefits:**
- Saves 6,000+ hours annually
 - Drastically improves test coverage and data visibility
 - Reduces human intervention and standardizes workflows
 - Eliminates the need for specialized product training.



The background is a dark, close-up photograph of a printed circuit board (PCB). A large, dark integrated circuit (chip) is the central focus, with white text printed on it: "88S9174-BLD2", "PFV6590A.1", and "1203 B1P". To the left of the chip, there are several electrolytic capacitors and other smaller components. The overall lighting is dim, highlighting the textures of the PCB and the components.

Navraj Kambo

Test and Automation Engineer (EIT)

Thank You!

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